

Table S1 Latitude, longitude, elevation (m a.s.l.), recorded air temperature (°C), number of flowering plant species (Plant rich), number of pollinator species (Poll rich), abundance of honeybees (Honeybee ab), and abundance of wild pollinators (Wild poll ab) for each sampling site

Site code	Latitude (WGS84)	Longitude (WGS84)	Elevation (m a.s.l.)	Temperature (°C)	Plant rich	Poll rich	Honeybee ab	Wild poll ab
A	45.587007	11.468313	60	27.67	20	31	5	111
AA	46.487789	12.035109	2005	25.29	26	22	768	73
AB	46.490416	12.066126	1982	23.24	28	12	109	16
AC	46.518712	12.007109	2090	22.36	34	19	145	39
AD	46.533594	11.986739	2111	27.05	25	13	282	43
AE	46.565153	12.243253	1658	22.37	21	30	34	65
AF	46.548764	12.261756	1500	26.72	19	16	22	29
AG	46.396688	12.104436	1450	25.43	27	49	274	119
AH	46.420795	12.105355	1780	20.95	28	30	31	150
AI	46.484398	11.831885	2008	19.63	22	14	2	93
AJ	45.82542	12.180076	175	27.99	13	27	44	63
AK	45.808239	12.111145	320	30.23	11	14	80	25
AL	46.485461	11.788397	1915	19.52	34	34	101	110
AM	46.549767	11.813354	2055	19.02	35	30	116	78
AN	46.339973	11.804519	2032	20.95	13	19	9	130
AO	46.195026	11.42058	1430	22.72	18	26	34	51
AP	46.348829	11.845274	1600	22.33	18	16	25	61
AQ	46.139193	11.472058	1250	24.49	16	24	149	332
AR	46.147208	11.771577	910	30.98	15	17	4	31
AS	46.172563	11.441852	2040	19.65	16	22	227	64
AT	45.880301	11.794101	1665	21.37	18	24	2	75
AU	45.755492	10.874539	1500	18.05	11	10	3	25
AV	45.756764	10.920759	640	24.63	10	9	108	29
AW	45.803576	12.049459	200	35.54	8	28	201	67
AX	45.868023	12.009987	170	30.91	10	13	24	45
AZ	45.694425	10.926014	130	32.84	10	12	55	16
B	45.583447	11.463828	150	27.37	22	29	25	119

C	45.747647	11.329589	670	24.04	19	35	113	109
D	45.755546	11.363854	630	24.65	17	29	64	69
E	45.711296	11.68899	76	30.60	15	32	302	94
F	45.760339	11.699603	106	36.89	12	25	508	73
G	45.595876	11.467499	150	37.89	22	22	19	121
H	45.622053	11.415189	165	33.93	16	19	19	103
I	45.862562	11.762562	1251	21.02	19	27	353	57
J	45.649502	11.7196	50	28.19	11	9	31	25
K	45.979623	12.730447	35	27.88	20	25	179	72
L	45.839134	11.736687	1000	25.27	15	17	550	51
M	45.848196	11.452165	943	26.37	29	29	119	93
N	45.824215	11.455581	1170	23.79	19	18	31	71
O	45.760665	11.39284	1215	19.51	22	34	21	119
P	46.137372	11.502687	1170	23.54	20	20	44	114
Q	46.121675	11.49488	920	27.36	19	22	66	61
R	46.154322	11.424568	1530	25.61	17	22	19	36
S	46.068526	11.431498	870	26.67	24	29	306	77
T	46.006465	11.404734	890	28.29	22	16	266	60
U	46.012846	11.432452	825	29.46	31	21	215	41
V	45.670237	11.056824	1430	24.46	32	35	5	191
W	46.196241	12.658885	870	21.16	25	29	150	112
X	45.758992	11.192283	1000	23.27	30	38	309	143
Y	45.769546	11.136029	860	24.26	20	34	115	109
Z	45.637685	11.720541	40	30.60	16	22	35	63

Table S2 Functional traits of flowering plant species: flower colour (six classes: blue, brown, pink/purple, red, white, yellow/orange), flower type (nine classes: bell/funnel/lip, brush, hidden disc, open disc, flag, flower head, NA, pollen, and stalk disc flower), and flower corolla length in mm

Plant species	Flower color	Flower type	Corolla (mm)
<i>Achillea clavennae</i>	white	flower head	6
<i>Achillea millefolium</i>	white	flower head	4.7
<i>Aconitum degenii</i>	blue	bell/funnel/lip	26
<i>Aconitum lycoctonum</i>	yellow/orange	bell/funnel/lip	25.5
<i>Aconitum napellus</i>	blue	bell/funnel/lip	11.5
<i>Adenostyles alpina</i>	pink/purple	stalk disc	13
<i>Aegopodium podagraria</i>	white	open disc	1
<i>Agrimonia eupatoria</i>	yellow/orange	pollen	0.05
<i>Allium carinatum</i>	pink/purple	hidden disc	5.5
<i>Allium schoenoprasum</i>	pink/purple	hidden disc	12
<i>Angelica sylvestris</i>	white	open disc	1
<i>Anthriscus sylvestris</i>	white	open disc	0.4
<i>Anthyllis vulneraria</i>	yellow/orange	flag	12
<i>Aristolochia clematitis</i>	yellow/orange	bell/funnel/lip	30
<i>Arnica montana</i>	yellow/orange	flower head	17.5
<i>Asperula cristata</i>	pink/purple	bell/funnel/lip	3
<i>Aster alpinus</i>	pink/purple	flower head	8
<i>Astragalus glycyphyllos</i>	yellow/orange	flag	7
<i>Astrantia major</i>	white	flower head	6
<i>Bellis perennis</i>	white	flower head	4
<i>Betonica alopecurus</i>	yellow/orange	bell/funnel/lip	10
<i>Buddleja davidii</i>	pink/purple	stalk disc	10
<i>Bupthalmum salicifolium</i>	yellow/orange	flower head	8
<i>Calluna vulgaris</i>	pink/purple	bell/funnel/lip	5
<i>Calystegia sepium</i>	white	bell/funnel/lip	33.3
<i>Campanula barbata</i>	pink/purple	bell/funnel/lip	16
<i>Campanula carnica</i>	pink/purple	bell/funnel/lip	11
<i>Campanula glomerata</i>	pink/purple	bell/funnel/lip	13

<i>Campanula patula</i>	pink/purple	bell/funnel/lip	11
<i>Campanula persicifolia</i>	pink/purple	bell/funnel/lip	18
<i>Campanula rapunculoides</i>	pink/purple	bell/funnel/lip	17
<i>Campanula rapunculus</i>	pink/purple	bell/funnel/lip	15
<i>Campanula rotundifolia</i>	pink/purple	bell/funnel/lip	15
<i>Campanula scheuchzeri</i>	pink/purple	bell/funnel/lip	15
<i>Campanula spicata</i>	pink/purple	bell/funnel/lip	11
<i>Campanula trachelium</i>	pink/purple	bell/funnel/lip	26.5
<i>Carduus defloratus</i>	pink/purple	flower head	17
<i>Carduus nutans</i>	pink/purple	flower head	12
<i>Carduus personata</i>	pink/purple	flower head	22
<i>Carum carvi</i>	white	open disc	0.3
<i>Centaurea jacea</i>	pink/purple	flower head	20
<i>Centaurea nervosa</i>	pink/purple	flower head	23.5
<i>Centaurea nigrescens</i>	pink/purple	flower head	12.2
<i>Centaurea scabiosa</i>	pink/purple	flower head	13.5
<i>Centaurea stoebe</i>	pink/purple	flower head	10
<i>Centaurea triumfetti</i>	blue	flower head	12
<i>Cerastium arvense</i>	white	hidden disc	4
<i>Cerastium holosteoides</i>	white	hidden disc	2
<i>Chaerophyllum hirsutum</i>	white	open disc	1
<i>Cichorium intybus</i>	blue	flower head	0.05
<i>Cirsium arvense</i>	pink/purple	flower head	16
<i>Cirsium erisithales</i>	yellow/orange	flower head	23.3
<i>Cirsium heterophyllum</i>	pink/purple	flower head	33
<i>Cirsium montanum</i>	pink/purple	flower head	7
<i>Cirsium oleraceum</i>	yellow/orange	flower head	23
<i>Clematis vitalba</i>	white	pollen	0.05
<i>Clinopodium acinos</i>	pink/purple	bell/funnel/lip	9
<i>Clinopodium alpinum</i>	pink/purple	bell/funnel/lip	9
<i>Clinopodium nepeta</i>	pink/purple	bell/funnel/lip	9.5
<i>Clinopodium vulgare</i>	pink/purple	bell/funnel/lip	11
<i>Convolvulus arvensis</i>	white	bell/funnel/lip	16.3

<i>Conyza canadensis</i>	white	flower head	5
<i>Crepis aurea</i>	yellow/orange	flower head	13.5
<i>Crepis biennis</i>	yellow/orange	flower head	8.5
<i>Crepis foetida</i>	yellow/orange	flower head	6
<i>Crepis paludosa</i>	yellow/orange	flower head	11
<i>Crepis vesicaria</i>	yellow/orange	flower head	8
<i>Cruciata laevipes</i>	yellow/orange	open disc	0.05
<i>Daucus carota</i>	white	open disc	0.05
<i>Delosperma</i> sp.	pink/purple	flower head	5
<i>Dianthus superbus</i>	pink/purple	stalk disc	30
<i>Diplotaxis tenuifolia</i>	yellow/orange	hidden disc	4
<i>Doronicum austriacum</i>	yellow/orange	flower head	4.5
<i>Dorycnium pentaphyllum</i>	white	flag	4
<i>Dryas octopetala</i>	white	hidden disc	0.05
<i>Echium vulgare</i>	pink/purple	bell/funnel/lip	9.3
<i>Epilobium angustifolium</i>	pink/purple	hidden disc	0.05
<i>Epilobium dodonaei</i>	pink/purple	hidden disc	0.05
<i>Epilobium hirsutum</i>	pink/purple	hidden disc	0.05
<i>Epilobium montanum</i>	pink/purple	hidden disc	7
<i>Erigeron annuus</i>	white	flower head	1
<i>Eupatorium cannabinum</i>	pink/purple	flower head	5
<i>Euphrasia rostkoviana</i>	white	bell/funnel/lip	10
<i>Euphrasia salisburgensis</i>	white	bell/funnel/lip	6
<i>Euphrasia</i> sp.	white	bell/funnel/lip	6.5
<i>Filipendula vulgaris</i>	white	pollen	0.05
<i>Fragaria vesca</i>	white	hidden disc	0.05
<i>Galeopsis pubescens</i>	pink/purple	bell/funnel/lip	17
<i>Galeopsis speciosa</i>	yellow/orange	bell/funnel/lip	22
<i>Galeopsis tetrahit</i>	pink/purple	bell/funnel/lip	13
<i>Galium lucidum</i>	white	open disc	0.5
<i>Galium mollugo</i>	white	open disc	1
<i>Galium saxatile</i>	white	open disc	1

<i>Galium verum</i>	yellow/orange	open disc	0.3
<i>Genista tinctoria</i>	yellow/orange	flag	7.7
<i>Gentiana cruciata</i>	blue	bell/funnel/lip	14.5
<i>Gentianella rhaetica</i>	pink/purple	bell/funnel/lip	17
<i>Geranium columbinum</i>	pink/purple	hidden disc	0.05
<i>Geranium molle</i>	pink/purple	hidden disc	0.05
<i>Geranium phaeum</i>	pink/purple	hidden disc	0.05
<i>Geranium pyrenaicum</i>	pink/purple	hidden disc	0.05
<i>Geranium robertianum</i>	pink/purple	hidden disc	6.5
<i>Geranium sylvaticum</i>	pink/purple	hidden disc	0.05
<i>Geum rivale</i>	red	bell/funnel/lip	7
<i>Gymandenia conopsea</i>	pink/purple	bell/funnel/lip	6
<i>Gypsophila repens</i>	white	bell/funnel/lip	7
<i>Hedysarum hedysaroides</i>	pink/purple	flag	8
<i>Helianthemum nummularium</i>	yellow/orange	pollen	0.05
<i>Heracleum sphondylium</i>	white	open disc	0.1
<i>Hieracium bifidum</i>	yellow/orange	flower head	13.3
<i>Hieracium glaucum</i>	yellow/orange	flower head	12
<i>Hieracium picroides</i>	yellow/orange	flower head	11
<i>Hieracium pilosella</i>	yellow/orange	flower head	11
<i>Hieracium</i> sp.	yellow/orange	flower head	8.5
<i>Hieracium valdepilosum</i>	yellow/orange	flower head	20
<i>Horminum pyrenaicum</i>	pink/purple	bell/funnel/lip	14
<i>Hypericum maculatum</i>	yellow/orange	pollen	0.05
<i>Hypericum perforatum</i>	yellow/orange	pollen	0.05
<i>Hypochaeris uniflora</i>	yellow/orange	flower head	22
<i>Impatiens glandulifera</i>	pink/purple	bell/funnel/lip	23
<i>Inula salicina</i>	yellow/orange	flower head	7.5
<i>Inula</i> sp.	yellow/orange	flower head	9
<i>Jacobaea alpina</i>	yellow/orange	flower head	7.5
<i>Knautia arvensis</i>	blue	flower head	7.3
<i>Knautia drymeia</i>	pink/purple	flower head	7

<i>Knautia longifolia</i>	pink/purple	flower head	9
<i>Lamium album</i>	white	bell/funnel/lip	9.5
<i>Lamium galeobdolon</i>	yellow/orange	bell/funnel/lip	9
<i>Lamium orvala</i>	pink/purple	bell/funnel/lip	14
<i>Lathyrus pratensis</i>	yellow/orange	flag	13.7
<i>Lathyrus sylvestris</i>	pink/purple	flag	12.5
<i>Leontodon hispidus</i>	yellow/orange	flower head	11.7
<i>Leucanthemum vulgare</i>	white	flower head	4.8
<i>Ligustrum lucidum</i>	white	bell/funnel/lip	3
<i>Ligustrum vulgare</i>	white	bell/funnel/lip	3
<i>Lilium bulbiferum</i>	yellow/orange	bell/funnel/lip	0.05
<i>Lilium martagon</i>	pink/purple	bell/funnel/lip	10
<i>Linaria alpina</i>	pink/purple	bell/funnel/lip	8
<i>Loncomelos brevistylus</i>	white	hidden disc	0.05
<i>Lotus corniculatus</i>	yellow/orange	flag	5.5
<i>Lupinus polyphyllus</i>	blue	flag	9
<i>Lychnis flos cuculi</i>	pink/purple	stalk disc	10.5
<i>Lysimachia arvensis</i>	yellow/orange	pollen	0.05
<i>Lysimachia vulgaris</i>	yellow/orange	pollen	0.05
<i>Lythrum salicaria</i>	pink/purple	bell/funnel/lip	7
<i>Malva sylvestris</i>	pink/purple	hidden disc	0.05
<i>Matricaria chamomilla</i>	white	flower head	7
<i>Medicago falcata</i>	yellow/orange	flag	5
<i>Medicago lupulina</i>	yellow/orange	flag	2.5
<i>Medicago sativa</i>	pink/purple	flag	6
<i>Melampyrum italicum</i>	yellow/orange	bell/funnel/lip	18
<i>Melampyrum sp.</i>	yellow/orange	bell/funnel/lip	10
<i>Melilotus albus</i>	white	flag	3.3
<i>Mentha arvensis</i>	pink/purple	bell/funnel/lip	4
<i>Mentha longifolia</i>	pink/purple	bell/funnel/lip	3
<i>Minuartia recurva</i>	white	hidden disc	0.05
<i>Myosotis sp.</i>	blue	stalk disc	2.3

<i>Myosoton aquaticum</i>	white	hidden disc	0.05
<i>Oenothera biennis</i>	yellow/orange	stalk disc	17
<i>Onobrychis montana</i>	pink/purple	flag	11.7
<i>Onobrychis viciifolia</i>	pink/purple	flag	11.5
<i>Ononis spinosa</i>	pink/purple	flag	10
<i>Origanum vulgare</i>	pink/purple	bell/funnel/lip	6
<i>Ornithogalum pyrenaicum</i>	white	hidden disc	0.05
<i>Ornithogalum umbellatum</i>	white	hidden disc	1
<i>Oxalis articulata</i>	pink/purple	hidden disc	3
<i>Oxytropis montana</i>	pink/purple	flag	11.5
<i>Papaver rhoeas</i>	red	pollen	0.05
<i>Parnassia palustris</i>	white	open disc	0.05
<i>Pastinaca sativa</i>	yellow/orange	open disc	1
<i>Pedicularis comosa</i>	yellow/orange	bell/funnel/lip	17
<i>Pedicularis palustris</i>	pink/purple	bell/funnel/lip	14
<i>Persicaria bistorta</i>	pink/purple	bell/funnel/lip	2
<i>Petrorhagia saxifraga</i>	pink/purple	bell/funnel/lip	4.7
<i>Phyteuma betonicifolium</i>	blue	NA	4.8
<i>Phyteuma orbiculare</i>	blue	NA	4
<i>Picris hieracioides</i>	yellow/orange	flower head	12.3
<i>Pimpinella major</i>	white	open disc	1.3
<i>Pimpinella saxifraga</i>	white	open disc	0.05
<i>Plantago lanceolata</i>	white	NA	0.05
<i>Plantago media</i>	white	NA	0.05
<i>Polygonum viviparum</i>	white	bell/funnel/lip	2
<i>Potentilla argentea</i>	yellow/orange	hidden disc	0.05
<i>Potentilla erecta</i>	yellow/orange	hidden disc	0.05
<i>Potentilla nitida</i>	pink/purple	hidden disc	0.05
<i>Potentilla reptans</i>	yellow/orange	hidden disc	0.05
<i>Prenanthes purpurea</i>	pink/purple	flower head	11.5
<i>Prunella grandiflora</i>	pink/purple	bell/funnel/lip	13
<i>Prunella laciniata</i>	white	bell/funnel/lip	6

<i>Prunella vulgaris</i>	pink/purple	bell/funnel/lip	8.4
<i>Pyrola minor</i>	white	bell/funnel/lip	0.05
<i>Ranunculus acris</i>	yellow/orange	hidden disc	0.05
<i>Ranunculus platanifolius</i>	white	hidden disc	0.05
<i>Ranunculus polyanthemophyllus</i>	yellow/orange	hidden disc	0.05
<i>Ranunculus repens</i>	yellow/orange	hidden disc	0.05
<i>Rhinanthus alectorolophus</i>	yellow/orange	bell/funnel/lip	22.5
<i>Rhinanthus freynii</i>	yellow/orange	bell/funnel/lip	18
<i>Rhinanthus minor</i>	yellow/orange	bell/funnel/lip	16
<i>Rhododendron hirsutum</i>	pink/purple	bell/funnel/lip	10.5
<i>Rorippa sylvestris</i>	yellow/orange	hidden disc	2
<i>Rosa canina</i>	pink/purple	pollen	0.05
<i>Rubus</i> sp.	white	hidden disc	0.05
<i>Salvia glutinosa</i>	yellow/orange	bell/funnel/lip	20
<i>Salvia pratensis</i>	blue	bell/funnel/lip	9
<i>Saponaria officinalis</i>	pink/purple	stalk disc	23
<i>Saxifraga rotundifolia</i>	white	open disc	0.05
<i>Scabiosa columbaria</i>	pink/purple	flower head	5.5
<i>Scabiosa gramuntia</i>	pink/purple	flower head	6.5
<i>Scabiosa lucida</i>	pink/purple	flower head	6
<i>Scabiosa triandra</i>	pink/purple	flower head	6.4
<i>Scrophularia nodosa</i>	red	bell/funnel/lip	6
<i>Securigera varia</i>	pink/purple	flag	11
<i>Sedum acre</i>	yellow/orange	hidden disc	1.75
<i>Sedum album</i>	white	hidden disc	1.5
<i>Senecio inaequidens</i>	yellow/orange	flower head	8
<i>Senecio nemorensis</i>	yellow/orange	flower head	13
<i>Senecio squalidus</i>	yellow/orange	flower head	10.3
<i>Sherardia arvensis</i>	pink/purple	bell/funnel/lip	2.5
<i>Silene alba</i>	white	stalk disc	16
<i>Silene dioica</i>	pink/purple	stalk disc	15.3
<i>Silene nutans</i>	white	stalk disc	13

<i>Silene saxifraga</i>	white	stalk disc	5.5
<i>Silene vulgaris</i>	white	stalk disc	18
<i>Solanum tuberosum</i>	white	pollen	2
<i>Solidago gigantea</i>	yellow/orange	flower head	6
<i>Solidago virgaurea</i>	yellow/orange	flower head	10.7
<i>Stachys alpina</i>	pink/purple	bell/funnel/lip	10
<i>Stachys officinalis</i>	pink/purple	bell/funnel/lip	9
<i>Stachys recta</i>	white	bell/funnel/lip	8
<i>Stachys sylvatica</i>	pink/purple	bell/funnel/lip	11
<i>Stellaria graminea</i>	white	hidden disc	0.05
<i>Stellaria nemorum</i>	white	hidden disc	2.5
<i>Symphytum officinale</i>	pink/purple	bell/funnel/lip	15
<i>Tanacetum corymbosum</i>	white	flower head	8
<i>Tanacetum vulgare</i>	yellow/orange	flower head	5
<i>Taraxacum officinale</i>	yellow/orange	flower head	8
<i>Teucrium chamaedrys</i>	pink/purple	bell/funnel/lip	8
<i>Thalictrum aquilegifolium</i>	pink/purple	brush	0.05
<i>Thalictrum lucidum</i>	yellow/orange	brush	0.05
<i>Thymus</i> sp.	pink/purple	bell/funnel/lip	3.5
<i>Torilis arvensis</i>	white	open disc	0.1
<i>Tragopogon pratensis</i>	yellow/orange	flower head	12.5
<i>Trifolium badium</i>	yellow/orange	flag	5.5
<i>Trifolium medium</i>	pink/purple	flag	13
<i>Trifolium montanum</i>	white	flag	8.5
<i>Trifolium pratense</i>	pink/purple	flag	11.3
<i>Trifolium repens</i>	white	flag	4
<i>Valeriana montana</i>	pink/purple	bell/funnel/lip	5
<i>Valeriana officinalis</i>	pink/purple	bell/funnel/lip	3.7
<i>Verbascum alpinum</i>	yellow/orange	bell/funnel/lip	8
<i>Verbascum chaixii</i>	yellow/orange	bell/funnel/lip	5
<i>Verbascum densiflorum</i>	yellow/orange	bell/funnel/lip	10
<i>Verbascum lychnitis</i>	yellow/orange	bell/funnel/lip	10

<i>Verbascum nigrum</i>	yellow/orange	bell/funnel/lip	3
<i>Verbena officinalis</i>	pink/purple	bell/funnel/lip	5.3
<i>Veronica anagallis-aquatica</i>	pink/purple	bell/funnel/lip	0.05
<i>Veronica chamaedrys</i>	blue	bell/funnel/lip	0.05
<i>Veronica persica</i>	blue	bell/funnel/lip	0.05
<i>Vicia cracca</i>	pink/purple	flag	7.6
<i>Vicia sepium</i>	pink/purple	flag	10
<i>Vicia sylvatica</i>	white	flag	13
<i>Vicia villosa</i>	pink/purple	flag	11

Table S3 Traits of pollinator species: taxonomic family, proboscis length in mm (mean value of the measured specimens), body size in mm (mean value of the measured specimens), and type of foraging range (C central-place forager, NC non-central-place forager)

Pollinator species	Family	Proboscis (mm)	Body (mm)	Foraging
<i>Abia fasciata</i>	Cimbicidae	0.8	12	NC
<i>Amegilla quadrifasciata</i>	Apidae	8.2	13	C
<i>Andrena aeneiventris</i>	Andrenidae	1.1	7	C
<i>Andrena alfkenella</i>	Andrenidae	0.95	5.75	C
<i>Andrena cineraria</i>	Andrenidae	2	12	C
<i>Andrena haemorrhoa</i>	Andrenidae	1.6	9	C
<i>Andrena hattorfiana</i>	Andrenidae	3.5	14	C
<i>Andrena intermedia</i>	Andrenidae	2.3	11	C
<i>Andrena labialis</i>	Andrenidae	2	11.5	C
<i>Andrena lathyri</i>	Andrenidae	2.7	10	C
<i>Andrena limata</i>	Andrenidae	1.85	11	C
<i>Andrena pandellei</i>	Andrenidae	1.95	9.5	C
<i>Andrena schencki</i>	Andrenidae	2.8	12	C
<i>Andrena</i> sp. 1	Andrenidae	1.6	9	C
<i>Andrena</i> sp. 2	Andrenidae	1.45	9	C
<i>Andrena</i> sp. 3	Andrenidae	1.5	8	C
<i>Andrena</i> sp. 4	Andrenidae	1.03	6.75	C
<i>Andrena</i> sp. 5	Andrenidae	1.15	10	C
<i>Andrena</i> sp. 6	Andrenidae	1.2	10	C
<i>Andrena</i> sp. 7	Andrenidae	1.25	8	C
<i>Andrena subopaca</i>	Andrenidae	1	5.5	C
<i>Andrena wilkella</i>	Andrenidae	1.95	9.5	C
<i>Anthidiellum strigatum</i>	Megachilidae	3.2	7	C
<i>Anthidium florentinum</i>	Megachilidae	4	13	C
<i>Anthidium oblongatum</i>	Megachilidae	2.7	7	C
<i>Anthidium punctatum</i>	Megachilidae	3.3	8	C
<i>Anthophora balneorum</i>	Apidae	12	15	C
<i>Anthophora furcata</i>	Apidae	8.25	11	C

<i>Anthophora plumipes</i>	Apidae	9	14	C
<i>Anthophora salviae</i>	Apidae	6	12	C
<i>Apis mellifera</i>	Apidae	5	12	C
<i>Argogorytes mystaceus</i>	Crabronidae	1.1	13	C
<i>Athalia rosae</i>	Tenthredinidae	0.5	6.5	NC
<i>Billaea triangulifera</i>	Tachinidae	2.3	10	NC
<i>Bombus argillaceus</i>	Apidae	10.7	15.67	C
<i>Bombus barbutellus</i>	Apidae	7.15	17	NC
<i>Bombus bohemicus</i>	Apidae	6.35	17.75	NC
<i>Bombus campestris</i>	Apidae	6.2	15	NC
<i>Bombus gerstaeckeri</i>	Apidae	16	18	C
<i>Bombus hortorum</i>	Apidae	12.53	16.83	C
<i>Bombus humilis</i>	Apidae	8.33	15	C
<i>Bombus hypnorum</i>	Apidae	7.25	13.5	C
<i>Bombus inexpectatus</i>	Apidae	7	15	NC
<i>Bombus lapidarius</i>	Apidae	7.27	15.33	C
<i>Bombus lucorum</i>	Apidae	5.5	16	C
<i>Bombus mendax</i>	Apidae	9.5	16	C
<i>Bombus mesomelas</i>	Apidae	8.5	15.5	C
<i>Bombus monticola</i>	Apidae	5.5	13	C
<i>Bombus pascuorum</i>	Apidae	8.25	15	C
<i>Bombus pratorum</i>	Apidae	7.67	14.67	C
<i>Bombus pyrenaeus</i>	Apidae	5.13	13.33	C
<i>Bombus ruderarius</i>	Apidae	7.13	13.25	C
<i>Bombus rupestris</i>	Apidae	7.4	18	NC
<i>Bombus sichelii</i>	Apidae	6.75	15.5	C
<i>Bombus soroeensis</i>	Apidae	6.35	17	C
<i>Bombus sylvarum</i>	Apidae	8.8	16	C
<i>Bombus sylvestris</i>	Apidae	6.5	16	NC
<i>Bombus terrestris</i>	Apidae	6.5	14.67	C
<i>Bombus vestalis</i>	Apidae	8.4	22	NC
<i>Bombus wurflenii</i>	Apidae	7.75	17	C
<i>Callicera aurata</i>	Syrphidae	1.5	12	NC

<i>Ceratina chalybaea</i>	Apidae	3.5	9.5	C
<i>Ceratina cucurbitina</i>	Apidae	3.5	8.5	C
<i>Ceratina cyanea</i>	Apidae	2.5	7.5	C
<i>Ceratina dallatorreana</i>	Apidae	2.6	6	C
<i>Ceratina gravidula</i>	Apidae	4.7	11	C
<i>Cerceris rubida</i>	Crabronidae	1.6	7.5	C
<i>Cerceris sabulosa</i>	Crabronidae	1.6	9	C
<i>Cheilosia aerea</i>	Syrphidae	1	8	NC
<i>Cheilosia albipila</i>	Syrphidae	1	10	NC
<i>Cheilosia antiqua</i>	Syrphidae	1	7	NC
<i>Cheilosia canicularis</i>	Syrphidae	1	12	NC
<i>Cheilosia frontalis</i>	Syrphidae	1	8.5	NC
<i>Cheilosia illustrata</i>	Syrphidae	1	10	NC
<i>Cheilosia laticornis</i>	Syrphidae	1	9	NC
<i>Cheilosia latifrons</i>	Syrphidae	1	9	NC
<i>Cheilosia longula</i>	Syrphidae	1	7.5	NC
<i>Cheilosia mutabilis</i>	Syrphidae	1	6.5	NC
<i>Cheilosia nigripes</i>	Syrphidae	1	7	NC
<i>Cheilosia pagana</i>	Syrphidae	1	6.5	NC
<i>Cheilosia personata</i>	Syrphidae	1	9.5	NC
<i>Cheilosia proxima</i>	Syrphidae	1	8	NC
<i>Cheilosia ranunculi</i>	Syrphidae	1	8	NC
<i>Cheilosia scutellata</i>	Syrphidae	1	8.5	NC
<i>Cheilosia soror</i>	Syrphidae	1	8.5	NC
<i>Cheilosia urbana</i>	Syrphidae	1	7	NC
<i>Cheilosia vernalis</i>	Syrphidae	1	6	NC
<i>Cheilosia vulpina</i>	Syrphidae	1	9	NC
<i>Chelostoma campanularum</i>	Megachilidae	2.9	6.5	C
<i>Chelostoma distinctum</i>	Megachilidae	1.8	7	C
<i>Chelostoma florissomne</i>	Megachilidae	2.3	9.5	C
<i>Chelostoma rapunculi</i>	Megachilidae	2.7	9	C
<i>Chrysosomoxys macrocercus</i>	Tachinidae	2.2	7.5	NC

<i>Chrysotoxum bicinctum</i>	Syrphidae	1	10.5	NC
<i>Chrysotoxum intermedium</i>	Syrphidae	1	12	NC
<i>Chrysotoxum vernale</i>	Syrphidae	1	11.5	NC
<i>Chrysotoxum verralli</i>	Syrphidae	1	11.5	NC
<i>Coelioxys conoidea</i>	Megachilidae	4.1	10	NC
<i>Conops flavipes</i>	Conopidae	4.2	11	NC
<i>Conops quadrifasciatus</i>	Conopidae	3.3	8.5	NC
<i>Corynis crassicornis</i>	Cimbicidae	0.9	8	NC
<i>Crossocerus cinxius</i>	Crabronidae	0.65	7	C
<i>Crossocerus leucostoma</i>	Crabronidae	0.6	8	C
<i>Cylindromyia brassicaria</i>	Tachinidae	2.4	9	NC
<i>Dasysyrphus albostrigatus</i>	Syrphidae	1	9	NC
<i>Didea alneti</i>	Syrphidae	1	14	NC
<i>Didea erratica</i>	Syrphidae	1	12.5	NC
<i>Dinera carinifrons</i>	Tachinidae	2.3	7.5	NC
<i>Dinera ferina</i>	Tachinidae	3.5	12	NC
<i>Dufourea alpina</i>	Halictidae	1.3	5.5	C
<i>Ectemnius borealis</i>	Crabronidae	0.95	7	C
<i>Ectemnius continuus</i>	Crabronidae	1.4	10	C
<i>Ectophasia crassipennis</i>	Tachinidae	3.3	9.75	NC
<i>Entomognathus brevis</i>	Crabronidae	0.4	4.5	C
<i>Epistrophe grossulariae</i>	Syrphidae	1	13	NC
<i>Epistrophe melanostoma</i>	Syrphidae	1	11	NC
<i>Epistrophe nitidicollis</i>	Syrphidae	1	11.5	NC
<i>Episyrphus balteatus</i>	Syrphidae	1	11.5	NC
<i>Eriozona syrphoides</i>	Syrphidae	1.5	14	NC
<i>Eristalinus taeniops</i>	Syrphidae	1.5	12.5	NC
<i>Eristalis arbustorum</i>	Syrphidae	1	10	NC
<i>Eristalis horticola</i>	Syrphidae	1.5	12	NC
<i>Eristalis interrupta</i>	Syrphidae	1	12	NC
<i>Eristalis jugorum</i>	Syrphidae	1.5	12.5	NC
<i>Eristalis pertinax</i>	Syrphidae	1.5	14	NC

<i>Eristalis rupium</i>	Syrphidae	1	11.5	NC
<i>Eristalis similis</i>	Syrphidae	1.5	14	NC
<i>Eristalis tenax</i>	Syrphidae	1.5	15	NC
<i>Erycia fatua</i>	Tachinidae	2.1	8.5	NC
<i>Eucera longicornis</i>	Apidae	6.25	14.5	C
<i>Eucera nigrescens</i>	Apidae	8.25	13	C
<i>Eulabidogaster setifacies</i>	Tachinidae	2.1	7	NC
<i>Eupeodes corollae</i>	Syrphidae	1	8	NC
<i>Eupeodes lapponicus</i>	Syrphidae	1	8	NC
<i>Eupeodes latifasciatus</i>	Syrphidae	1	10	NC
<i>Eupeodes luniger</i>	Syrphidae	1	9.5	NC
<i>Eupeodes tirolensis</i>	Syrphidae	1	10	NC
<i>Exorista rustica</i>	Tachinidae	2.2	12	NC
<i>Exorista tubulosa</i>	Tachinidae	2.4	8.5	NC
<i>Gorytes quinquecinctus</i>	Crabronidae	0.7	10	C
<i>Gymnosoma clavatum</i>	Tachinidae	1.9	7	NC
<i>Gymnosoma nitens</i>	Tachinidae	1.7	5	NC
<i>Gymnosoma rotundatum</i>	Tachinidae	2.4	7.75	NC
<i>Gymnosoma</i> sp.	Tachinidae	2	7	NC
<i>Halictus compressus</i>	Halictidae	2.55	9.5	C
<i>Halictus langobardicus</i>	Halictidae	2.1	8.5	C
<i>Halictus maculatus</i>	Halictidae	1.6	8.5	C
<i>Halictus rubicundus</i>	Halictidae	2.23	10.5	C
<i>Halictus scabiosae</i>	Halictidae	3.65	13	C
<i>Halictus simplex</i>	Halictidae	2.75	9.75	C
<i>Halictus subauratus</i>	Halictidae	1.85	7	C
<i>Helophilus pendulus</i>	Syrphidae	1	12	NC
<i>Helophilus trivittatus</i>	Syrphidae	1	15.5	NC
<i>Heriades rubicola</i>	Megachilidae	1.8	6	C
<i>Heriades truncorum</i>	Megachilidae	2.2	8	C
<i>Hoplitis adunca</i>	Megachilidae	4.5	9	C
<i>Hoplitis</i> sp. 1	Megachilidae	3.9	7.5	C

<i>Hoplitis villosa</i>	Megachilidae	4.8	12	C
<i>Hyalurgus cruciger</i>	Tachinidae	1.1	5.5	NC
<i>Hylaeus brevicornis</i>	Colletidae	0.75	4.75	C
<i>Hylaeus communis</i>	Colletidae	0.75	4.75	C
<i>Hylaeus confusus</i>	Colletidae	1	5.5	C
<i>Hylaeus gibbus</i>	Colletidae	1.1	6.75	C
<i>Hylaeus hyalinatus</i>	Colletidae	0.8	5.3	C
<i>Hylaeus imparilis</i>	Colletidae	0.55	4	C
<i>Hylaeus punctatus</i>	Colletidae	0.7	5	C
<i>Hylaeus taeniolatus</i>	Colletidae	0.6	4	C
<i>Hylaeus tyrolensis</i>	Colletidae	0.6	5	C
<i>Hylaeus variegatus</i>	Colletidae	1	6.25	C
<i>Lasioglossum albipes</i>	Halictidae	1.5	8	C
<i>Lasioglossum angusticeps</i>	Halictidae	1.8	7	C
<i>Lasioglossum calceatum</i>	Halictidae	2.2	10	C
<i>Lasioglossum discum</i>	Halictidae	2.6	8.75	C
<i>Lasioglossum fulvicorne</i>	Halictidae	1.3	7.5	C
<i>Lasioglossum glabriusculum</i>	Halictidae	1	4.5	C
<i>Lasioglossum lativentre</i>	Halictidae	1.7	8	C
<i>Lasioglossum leucozonium</i>	Halictidae	2.25	8	C
<i>Lasioglossum malachurum</i>	Halictidae	1.8	9	C
<i>Lasioglossum morio</i>	Halictidae	1.1	7	C
<i>Lasioglossum nigripes</i>	Halictidae	2.45	9.75	C
<i>Lasioglossum parvulum</i>	Halictidae	1.3	6.5	C
<i>Lasioglossum politum</i>	Halictidae	0.93	5.25	C
<i>Lasioglossum punctatissimum</i>	Halictidae	1.3	6	C
<i>Lasioglossum pygmaeum</i>	Halictidae	1	6	C
<i>Lasioglossum rufitarse</i>	Halictidae	1.3	6	C
<i>Lasioglossum</i> sp. 1	Halictidae	1.2	6.5	C
<i>Lasioglossum</i> sp. 2	Halictidae	2.5	8	C
<i>Lasioglossum villosulum</i>	Halictidae	1.65	6	C
<i>Lasioglossum zonulum</i>	Halictidae	2.5	8.5	C

<i>Leucostoma simplex</i>	Tachinidae	2.2	7	NC
<i>Leucozona lucorum</i>	Syrphidae	1	11.5	NC
<i>Lindenius albilabris</i>	Crabronidae	0.7	6	C
<i>Linnaemya impudica</i>	Tachinidae	4.3	13	NC
<i>Linnaemya lithosiophaga</i>	Tachinidae	1.7	7	NC
<i>Linnaemya picta</i>	Tachinidae	3.3	12	NC
<i>Linnaemya zachvatkini</i>	Tachinidae	3	12	NC
<i>Macrophya montana</i>	Tenthredinidae	0.95	11.25	NC
<i>Macropis europaea</i>	Melittidae	1.2	9	C
<i>Masistylum arcuatum</i>	Tachinidae	2.8	8.25	NC
<i>Megachile circumcincta</i>	Megachilidae	3.2	12	C
<i>Megachile lagopoda</i>	Megachilidae	6.5	18	C
<i>Megachile leachella</i>	Megachilidae	3.7	9	C
<i>Megachile ligniseca</i>	Megachilidae	4.5	14	C
<i>Megachile melanopyga</i>	Megachilidae	3.6	11	C
<i>Megachile nigriventris</i>	Megachilidae	6.2	15	C
<i>Megachile pilidens</i>	Megachilidae	3.7	8.75	C
<i>Megachile sculpturalis</i>	Megachilidae	5.65	20	C
<i>Megachile</i> sp. 1	Megachilidae	4.8	15	C
<i>Megachile</i> sp. 2	Megachilidae	4.1	12.5	C
<i>Megachile versicolor</i>	Megachilidae	4	9	C
<i>Megachile willughbiella</i>	Megachilidae	4.45	12.25	C
<i>Megalodontes</i> sp. 1	Megalodontesidae	2.3	12	NC
<i>Megalodontes</i> sp. 2	Megalodontesidae	2.2	12	NC
<i>Melangyna compositarum</i>	Syrphidae	1	10	NC
<i>Melanogaster nuda</i>	Syrphidae	0.6	5.5	NC
<i>Melanostoma mellinum</i>	Syrphidae	0.6	6	NC
<i>Melanostoma scalare</i>	Syrphidae	0.6	8.5	NC
<i>Meliscaeva auricollis</i>	Syrphidae	1	9.5	NC
<i>Meliscaeva cinctella</i>	Syrphidae	1	10	NC
<i>Merodon aeneus</i>	Syrphidae	1	8.5	NC

<i>Merodon cinereus</i>	Syrphidae	1	9	NC
<i>Merodon costans</i>	Syrphidae	1	11	NC
<i>Merodon equestris</i>	Syrphidae	1	13	NC
<i>Merodon funestus</i>	Syrphidae	1	9	NC
<i>Merodon</i> sp.	Syrphidae	1	10	NC
<i>Mintho rufiventris</i>	Tachinidae	1.4	7	NC
<i>Myathropa florea</i>	Syrphidae	1	12	NC
<i>Neoascia podagrica</i>	Syrphidae	0.6	5.5	NC
<i>Nomada armata</i>	Apidae	3.5	9	NC
<i>Nomada flavopicta</i>	Apidae	3.1	8	NC
<i>Nomada sexfasciata</i>	Apidae	4.7	11	NC
<i>Nowickia ferox</i>	Tachinidae	4.45	14.5	NC
<i>Nowickia marklini</i>	Tachinidae	4.3	13	NC
<i>Osmia aurulenta</i>	Megachilidae	4	12	C
<i>Osmia bicolor</i>	Megachilidae	5.2	11.5	C
<i>Osmia caerulea</i>	Megachilidae	4.9	7.5	C
<i>Osmia leaiana</i>	Megachilidae	4	9	C
<i>Osmia rufohirta</i>	Megachilidae	4	7	C
<i>Oxybelus mucronatus</i>	Crabronidae	0.8	5	C
<i>Oxybelus trispinosus</i>	Crabronidae	1	5.5	C
<i>Panurginus montanus</i>	Andrenidae	1.4	7.5	C
<i>Panzeria vivida</i>	Tachinidae	2.7	8.5	NC
<i>Paragus bicolor</i>	Syrphidae	0.6	6.5	NC
<i>Paragus constrictus</i>	Syrphidae	0.6	5	NC
<i>Paragus haemorrhous</i>	Syrphidae	0.6	6.5	NC
<i>Paragus</i> sp.	Syrphidae	0.6	6	NC
<i>Paragus tibialis</i>	Syrphidae	0.6	6.5	NC
<i>Parasyrphus lineolus</i>	Syrphidae	1	9	NC
<i>Peleteria iavana</i>	Tachinidae	2.6	9	NC
<i>Phasia aurulans</i>	Tachinidae	2.1	6	NC
<i>Phasia obesa</i>	Tachinidae	1.6	5	NC
<i>Physocephala rufipes</i>	Conopidae	5	13	NC

<i>Physocephala vittata</i>	Conopidae	4.7	10.5	NC
<i>Pipiza austriaca</i>	Syrphidae	0.6	8.5	NC
<i>Pipiza lugubris</i>	Syrphidae	0.6	8	NC
<i>Pipiza</i> sp.	Syrphidae	0.6	8	NC
<i>Pipizella divicoi</i>	Syrphidae	0.6	6.5	NC
<i>Pipizella</i> sp.	Syrphidae	0.6	6	NC
<i>Pipizella viduata</i>	Syrphidae	0.6	6.5	NC
<i>Platycheirus albimanus</i>	Syrphidae	0.6	8.5	NC
<i>Platycheirus scutatus</i>	Syrphidae	0.6	8.5	NC
<i>Platycheirus</i> sp.	Syrphidae	0.6	8.5	NC
<i>Platycheirus tarsalis</i>	Syrphidae	0.6	8.5	NC
<i>Polistes associus</i>	Vespidae	1.5	14	C
<i>Polistes biglumis</i>	Vespidae	1.65	13	C
<i>Polistes dominula</i>	Vespidae	1.6	14	C
<i>Polistes gallicus</i>	Vespidae	1.6	14	C
<i>Polistes nimpha</i>	Vespidae	1.4	13.5	C
<i>Polistes semenowi</i>	Vespidae	1.5	13	NC
<i>Prosenia siberita</i>	Tachinidae	5.9	8	NC
<i>Psenulus pallipes</i>	Crabronidae	0.5	6.5	C
<i>Pseudapis diversipes</i>	Halictidae	2.3	8.25	C
<i>Pseudoanthidium scapulare</i>	Megachilidae	3.75	6.5	C
<i>Rhingia campestris</i>	Syrphidae	3.5	9.5	NC
<i>Rhodanthidium septemdentatum</i>	Megachilidae	4.5	12.5	C
<i>Rhogogaster picta</i>	Tenthredinidae	0.6	12	NC
<i>Scaeva pyrastris</i>	Syrphidae	1.5	12.5	NC
<i>Scaeva selenitica</i>	Syrphidae	1.5	13.5	NC
<i>Scolia hirta</i>	Scoliidae	4.1	19	NC
<i>Sicus ferrugineus</i>	Conopidae	5.8	11.25	NC
<i>Siphona flavifrons</i>	Tachinidae	3.7	5.5	NC
<i>Siphona geniculata</i>	Tachinidae	3.05	5	NC
<i>Solieria vacua</i>	Tachinidae	1.75	6.75	NC
<i>Sphaerophoria infuscata</i>	Syrphidae	0.6	8.5	NC

<i>Sphaerophoria interrupta</i>	Syrphidae	0.6	8.5	NC
<i>Sphaerophoria scripta</i>	Syrphidae	0.6	11.5	NC
<i>Sphaerophoria</i> sp.	Syrphidae	0.6	9.5	NC
<i>Sphaerophoria taeniata</i>	Syrphidae	0.6	9	NC
<i>Sphecodes gibbus</i>	Halictidae	1.7	8	NC
<i>Sphecodes monilicornis</i>	Halictidae	2.1	9	NC
<i>Sphecodes pellucidus</i>	Halictidae	1.5	8	NC
<i>Sphecodes schencki</i>	Halictidae	1.7	8	NC
<i>Sphegina clunipes</i>	Syrphidae	0.6	6.5	NC
<i>Sphegina</i> sp.	Syrphidae	0.6	6.5	NC
<i>Stelis punctulatissima</i>	Megachilidae	3.1	8.5	NC
<i>Strongygaster globula</i>	Tachinidae	1	6	NC
<i>Syritta pipiens</i>	Syrphidae	0.6	8	NC
<i>Syrphus ribesii</i>	Syrphidae	1	11	NC
<i>Syrphus torvus</i>	Syrphidae	1	11.5	NC
<i>Syrphus vitripennis</i>	Syrphidae	1	9.5	NC
<i>Tachina fera</i>	Tachinidae	4.8	14	NC
<i>Tachina magnicornis</i>	Tachinidae	3.45	12.5	NC
<i>Tenthredo arcuata</i>	Tenthredinidae	1	10.5	NC
<i>Tenthredo crassa</i>	Tenthredinidae	1.05	13	NC
<i>Tenthredo koehleri</i>	Tenthredinidae	1.7	11.5	NC
<i>Tenthredo olivacea</i>	Tenthredinidae	0.9	11	NC
<i>Tenthredo rubricoxis</i>	Tenthredinidae	0.8	12	NC
<i>Tenthredo</i> sp. 1	Tenthredinidae	0.9	10.5	NC
<i>Tenthredo</i> sp. 2	Tenthredinidae	1	11	NC
<i>Tenthredo zonula</i>	Tenthredinidae	0.68	8.5	NC
<i>Tenthredopsis</i> sp. 1	Tenthredinidae	0.9	13	NC
<i>Tenthredopsis tischbeinii</i>	Tenthredinidae	0.9	11	NC
<i>Tetraloniella dentata</i>	Apidae	4.3	12	C
<i>Tetraloniella salicariae</i>	Apidae	3	8	C
<i>Thecophora atra</i>	Conopidae	3	4.8	NC
<i>Thecophora distincta</i>	Conopidae	3.4	5.2	NC

<i>Trachusa byssina</i>	Megachilidae	4.45	10.5	C
<i>Trichopoda pennipes</i>	Tachinidae	2.4	8.5	NC
<i>Volucella bombylans</i>	Syrphidae	1.5	13	NC
<i>Volucella pellucens</i>	Syrphidae	1.5	15.5	NC
<i>Xylocopa iris</i>	Apidae	6	19	C
<i>Xylocopa violacea</i>	Apidae	7.85	22.5	C
<i>Xylota jakutorum</i>	Syrphidae	1	11	NC
<i>Zodion cinereum</i>	Conopidae	3.4	6	NC
<i>Zophomyia temula</i>	Tachinidae	2.7	11	NC

Table S4 Traits of pollinator families: number of collected specimens (N), proboscis length in mm (mean and SD values of the measured specimens), body size in mm (mean and SD values of the measured specimens), and percentage of central and non-central forager species (*C* central-place foragers, *NC* non-central-place foragers)

Pollinator family	N	Proboscis (mm)	Body (mm)	C species (%)	NC species (%)
Andrenidae	86	1.73 (0.67)	9.33 (2.17)	100	-
Apidae	8045	6.88 (2.66)	14.1 (3.56)	78	22
Cimbicidae	2	0.85 (0.07)	10 (2.83)	-	100
Colletidae	37	0.79 (0.19)	5.13 (0.88)	100	-
Conopidae	43	4.1 (0.99)	8.78 (3.12)	-	100
Crabronidae	40	0.92 (0.4)	7.62 (2.38)	100	-
Halictidae	258	1.84 (0.62)	7.87 (1.73)	88	12
Megachilidae	116	3.9 (1.11)	10.31 (3.32)	94	6
Megalodontesidae	3	2.25 (0.07)	12 (0)	-	100
Melittidae	8	1.2 (NA)	9 (NA)	100	-
Scoliidae	4	4.1 (NA)	19 (NA)	-	100
Syrphidae	2000	0.98 (0.38)	9.66 (2.49)	-	100
Tachinidae	102	2.68 (1.07)	8.74 (2.72)	-	100
Tenthredinidae	81	0.91 (0.29)	10.9 (1.76)	-	100
Vespidae	16	1.54 (0.09)	13.58 (0.49)	83	17

Table S5 Summary of the six global models. Abbreviated explanatory variables are honeybee abundance (*Apis*, ln-transformed), temperature (Temp), standardized functional richness of plant community (FRic), functional dispersion of plant community (FDis), trait similarity between wild pollinator community and the honeybee (TSim), proboscis length category (Prob), body size category (Body), type of foraging range (For), and taxonomic family (Fam). The continuous explanatory variables were scaled to mean 0 and standard deviation 1 to make slopes comparable

	Type of model	Response variable	Explanatory variables	Random effect
Model 1	Linear model	CWM resource overlap between wild pollinators and the honeybee (one value per network)	$Apis + Temp + FRic + TSim + Apis \times FRic + Apis \times TSim + FRic \times TSim + Apis \times FRic \times TSim$	NA
Model 2	Linear model	CWM resource overlap between wild pollinators and the honeybee (one value per network)	$Apis + Temp + FDis + TSim + Apis \times FDis + Apis \times TSim + FDis \times TSim + Apis \times FDis \times TSim$	NA
Model 3	Linear mixed-effect model	CWM resource overlap between wild pollinators and the honeybee (one value per network per trait category, i.e., proboscis shorter, similar, and longer than the honeybee)	$Apis \times Prob + Temp$	Network
Model 4	Linear mixed-effect model	CWM resource overlap between wild pollinators and the honeybee (one value per network per trait category, i.e., smaller, similar, and larger than the honeybee)	$Apis \times Body + Temp$	Network
Model 5	Linear mixed-effect model	CWM resource overlap between wild pollinators and the honeybee (one value per network per trait category, i.e., central forager and non-central forager)	$Apis \times For + Temp$	Network

Model 6	Linear mixed-effect model	CWM resource overlap between wild pollinators and the honeybee (one value per network per trait category, i.e., Andrenidae, Apidae, Colletidae, Conopidae, Crabronidae, Halictidae, Megachilidae, other families, Syrphidae, Tachinidae, Tenthredinidae, and Vespidae)	$Apis \times \text{Fam} + \text{Temp}$	Network
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Table S6 Results of the multi-model inference analysis testing the effects on CWM resource overlap of honeybee abundance (*Apis*, ln-transformed), temperature (Temp), standardized functional richness of plant community (FRic), trait similarity between wild pollinator community and the honeybee (TSim), and the following interactions: *Apis* × FRic, *Apis* × TSim, FRic × TSim, and *Apis* × FRic × TSim. The table reports the estimates for each variable, the ΔAICc , the R^2 , and the Akaike weight (w) for each model with $\Delta\text{AICc} < 6$. All the explanatory variables were scaled to mean 0 and standard deviation 1

Ranking	Intercept	<i>Apis</i>	Temp	FRic	TSim	<i>Apis</i> × FRic	<i>Apis</i> × TSim	FRic × TSim	<i>Apis</i> × FRic × TSim	ΔAICc	R^2	w
1	-1.471	-0.087	-	-0.010	0.143	-0.193	-	-	-	0.000	0.290	0.286
2	-1.474	-0.084	-	-0.008	0.155	-0.201	-0.058	-	-	1.917	0.300	0.110
3	-1.458	-0.087	-	0.000	0.159	-0.190	-	0.043	-	2.290	0.295	0.091
4	-1.472	-0.089	0.007	-0.007	0.144	-0.191	-	-	-	2.682	0.290	0.075
5	-1.472	-0.087	-	-0.056	-	-0.189	-	-	-	3.373	0.202	0.053
6	-1.517	-	-	-	0.158	-	-	-	-	3.511	0.120	0.049
7	-1.517	-0.084	-	-	0.152	-	-	-	-	3.884	0.154	0.041
8	-1.458	-0.083	-	0.006	0.177	-0.199	-0.069	0.055	-	4.065	0.310	0.037
9	-1.474	-0.084	0.000	-0.007	0.155	-0.201	-0.058	-	-	4.741	0.300	0.027
10	-1.517	-	-	-0.061	0.139	-	-	-	-	4.931	0.136	0.024
11	-1.459	-0.089	0.007	0.003	0.160	-0.188	-	0.043	-	5.099	0.295	0.022
12	-1.517	-	0.048	-	0.159	-	-	-	-	5.226	0.131	0.021
13	-1.517	-0.093	0.061	-	0.152	-	-	-	-	5.270	0.172	0.021
14	-1.517	-0.075	-	-0.043	0.139	-	-	-	-	5.890	0.161	0.015
15	-1.471	-0.084	-0.015	-0.062	-	-0.194	-	-	-	5.894	0.203	0.015

Table S7 Results of the multi-model inference analysis testing the effects on CWM resource overlap between wild pollinators and the honeybee of honeybee abundance (*Apis*, ln-transformed), temperature (Temp), functional dispersion of plant community (FDis), trait similarity between wild pollinator community and the honeybee (TSim), and the following interactions: *Apis* × FDis, *Apis* × TSim, FDis × TSim, and *Apis* × FDis × TSim. The table reports the estimates for each variable, the ΔAICc , the R^2 , and the Akaike weight (w) for each model with $\Delta\text{AICc} < 6$. All the explanatory variables were scaled to mean 0 and standard deviation 1

Ranking	Intercept	<i>Apis</i>	Temp	FDis	TSim	<i>Apis</i> × FDis	<i>Apis</i> × TSim	FDis × TSim	<i>Apis</i> × FDis × TSim	ΔAICc	R^2	w
1	-1.517	-	-	-	0.171	-	-	-	-	0.000	0.141	0.144
2	-1.517	-0.082	-	-	0.165	-	-	-	-	0.436	0.173	0.116
3	-1.517	-	-	-0.082	0.143	-	-	-	-	0.635	0.169	0.105
4	-1.517	-0.087	-	-0.088	0.134	-	-	-	-	0.842	0.205	0.095
5	-1.517	-	0.038	-	0.170	-	-	-	-	1.944	0.148	0.055
6	-1.517	-0.090	0.052	-	0.161	-	-	-	-	2.128	0.185	0.050
7	-1.519	-0.081	-	-	0.168	-	-0.023	-	-	2.793	0.174	0.036
8	-1.527	-	-	-0.067	0.140	-	-	-0.030	-	2.857	0.173	0.035
9	-1.519	-0.079	-	-0.084	0.135	-0.059	-	-	-	2.967	0.212	0.033
10	-1.517	-0.100	-	-0.134	-	-	-	-	-	2.981	0.130	0.033
11	-1.517	-	0.021	-0.077	0.144	-	-	-	-	2.982	0.171	0.032
12	-1.517	-0.092	0.034	-0.080	0.135	-	-	-	-	3.097	0.210	0.031
13	-1.527	-0.088	-	-0.072	0.130	-	-	-0.031	-	3.135	0.210	0.030
14	-1.518	-0.086	-	-0.087	0.137	-	-0.019	-	-	3.344	0.206	0.027

15	-1.517	-	-	-0.131	-	-	-	-	-	3.344	0.083	0.027
16	-1.518	-0.089	0.050	-	0.164	-	-0.016	-	-	4.650	0.186	0.014
17	-1.518	-0.092	-	-0.131	-	-0.056	-	-	-	5.077	0.136	0.011
18	-1.517	-0.104	0.030	-0.127	-	-	-	-	-	5.205	0.134	0.011
19	-1.528	-	0.026	-0.058	0.140	-	-	-0.034	-	5.245	0.176	0.010
20	-1.530	-0.093	0.040	-0.059	0.130	-	-	-0.039	-	5.377	0.217	0.010
21	-1.529	-0.079	-	-0.069	0.131	-0.058	-	-0.031	-	5.390	0.216	0.010
22	-1.517	-0.096	-	-	-	-	-	-	-	5.456	0.044	0.009
23	-1.522	-0.076	-	-0.082	0.140	-0.069	-0.032	-	-	5.464	0.215	0.009
24	-1.517	-	-	-	-	-	-	-	-	5.477	0.000	0.009
25	-1.518	-0.083	0.023	-0.079	0.135	-0.047	-	-	-	5.535	0.214	0.009
26	-1.517	-	0.015	-0.128	-	-	-	-	-	5.646	0.084	0.009
27	-1.530	-0.087	-	-0.070	0.134	-	-0.025	-0.034	-	5.710	0.212	0.008
28	-1.518	-0.091	0.032	-0.080	0.137	-	-0.015	-	-	5.747	0.211	0.008

Table S8 Results of the multi-model inference analysis for single traits of pollinators, testing the effects on CWM resource overlap between wild pollinators and the honeybee of honeybee abundance (*Apis*, ln-transformed), temperature (Temp), and **a** proboscis length (Prob) and the interaction between honeybee abundance and proboscis length (*Apis* × Prob), **b** body size (Body) and the interaction between honeybee abundance and body size (*Apis* × Body), **c** type of foraging range (For) and the interaction between honeybee abundance and type of foraging range (*Apis* × For), and **d** taxonomic family (Fam) and the interaction between honeybee abundance and taxonomic family (*Apis* × Fam). The table reports the estimate for each continuous variable or the presence of each categorical variable in the model, the ΔAICc , the R^2 , and the Akaike weight (w) for each model of the set with $\Delta\text{AICc} < 6$. All the continuous explanatory variables were scaled to mean 0 and standard deviation 1

(a) Proboscis length								
Ranking	Intercept	<i>Apis</i>	Temp	Prob	<i>Apis</i> × Prob	ΔAICc	R^2	w
1	-1.675	-	-	+	-	0.000	0.117	0.487
2	-1.674	-0.032	-	+	-	1.918	0.119	0.187
3	-1.676	-	0.025	+	-	2.017	0.118	0.178
4	-1.675	-0.036	0.030	+	-	3.911	0.120	0.069
5	-1.672	-0.099	-	+	+	4.893	0.128	0.042
6	-1.673	-0.104	0.032	+	+	6.916	0.129	0.015
(b) Body size category								
Ranking	Intercept	<i>Apis</i>	Temp	Body	<i>Apis</i> × Body	ΔAICc	R^2	w
1	-1.987	-	0.087	+	-	0.000	0.249	0.339
2	-1.986	-	-	+	-	0.533	0.236	0.260
3	-1.985	-0.061	0.096	+	-	0.811	0.257	0.226
4	-1.985	-0.048	-	+	-	1.899	0.240	0.131
5	-1.987	-0.047	0.096	+	+	5.025	0.258	0.027
6	-1.986	-0.034	-	+	+	6.085	0.241	0.016
(c) Type of foraging range								
Ranking	Intercept	<i>Apis</i>	Temp	For	<i>Apis</i> × For	ΔAICc	R^2	w
1	-1.394	-0.092	-	+	-	0.000	0.135	0.343
2	-1.394	-	-	+	-	0.653	0.110	0.248
3	-1.395	-0.098	0.041	+	-	1.674	0.140	0.149
4	-1.394	-0.085	-	+	+	2.236	0.135	0.112
5	-1.395	-	0.027	+	-	2.624	0.112	0.092
6	-1.395	-0.091	0.041	+	+	3.959	0.140	0.047
(d) Taxonomic family								
Ranking	Intercept	<i>Apis</i>	Temp	Fam	<i>Apis</i> × Fam	ΔAICc	R^2	w
1	-1.987	-0.074	0.081	+	-	0.000	0.239	0.393
2	-1.993	-	0.071	+	-	1.049	0.230	0.232

3	-1.988	-0.065	-	+	-	1.401	0.230	0.195
4	-1.992	-	-	+	-	1.583	0.223	0.178

Fig. S1 Map of the 51 sampling sites. Map credit: © OpenStreetMap contributors

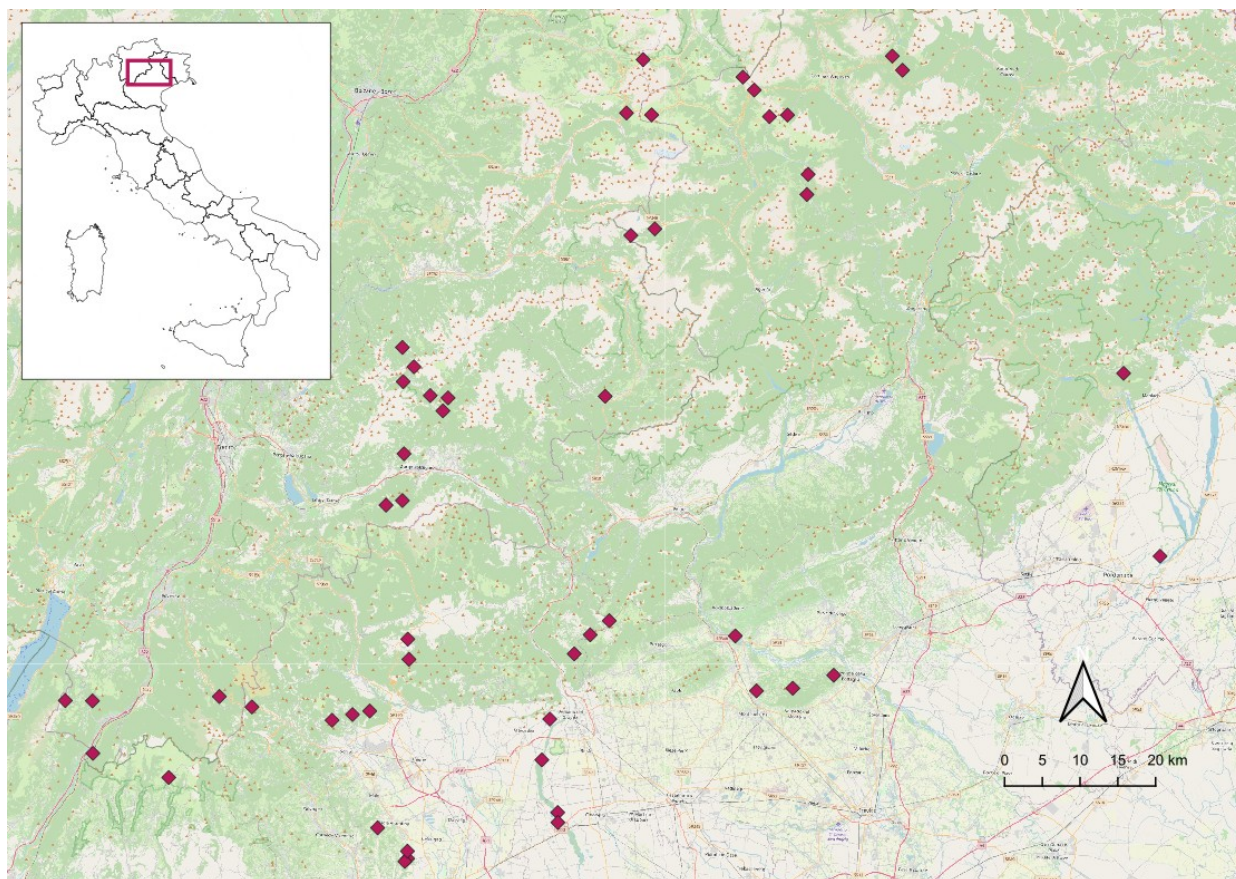


Fig. S2 Plots showing the relationships between functional traits of pollinators, i.e., **a** proboscis length and body size, **b** type of foraging range and proboscis length, and **c** type of foraging range and body size. In all plots, proboscis length and body size were log-transformed

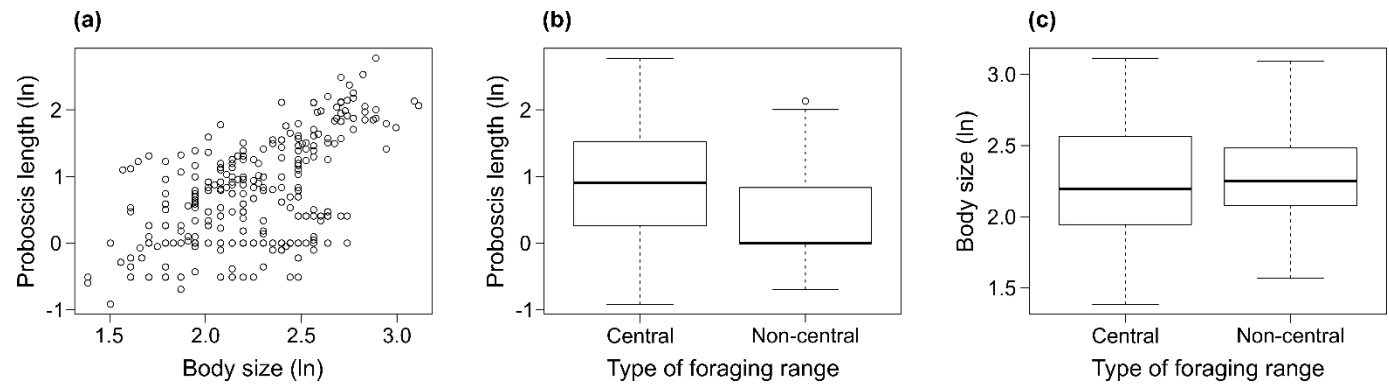
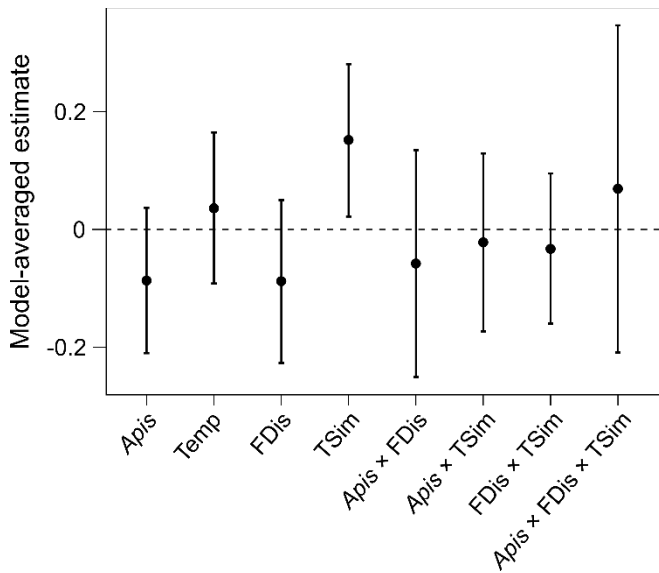


Fig. S3 Model estimates from the model-averaging procedure based on the set of models with all functional traits of both plants and pollinators. Explanatory variables of the global model are honeybee abundance (*Apis*, ln-transformed), temperature (Temp), functional dispersion of plant community (FDis), trait similarity between wild pollinator community and the honeybee (TSim), and the following interactions: *Apis* × FDis, *Apis* × TSim, FDis × TSim, and *Apis* × FDis × TSim. All explanatory variables were scaled to mean 0 and standard deviation 1. Dots indicate the model estimated means, while error bars indicate the 95% confidence intervals for the expected value of the variable. The variable effect is supported when the confidence intervals do not include zero



Additional references for pollinator functional traits

Alford DV (1975) Bumblebees. Davis-Poynter, London

Amiet F, Müller A, Neumeyer R (2014) Fauna Helvetica 4, Apidae 2 – *Colletes*, *Dufourea*, *Hylaeus*, *Nomia*, *Nomioides*, *Rhopitoides*, *Rophites*, *Sphecodes*, *Systropha*. Centre Suisse de Cartographie de la Faune & Schweizerische Entomologische Gesellschaft, Neuchâtel

Amiet F, Herrmann M, Müller A, et al (2001) Fauna Helvetica 6, Apidae 3 – *Halictus*, *Lasioglossum*. Centre Suisse de Cartographie de la Faune & Schweizerische Entomologische Gesellschaft, Neuchâtel

Amiet F, Herrmann M, Müller A, et al (2004) Fauna Helvetica 9, Apidae 4 – *Anthidium*, *Chelostoma*, *Coelioxys*, *Dioxys*, *Heriades*, *Lithurgus*, *Megachile*, *Osmia*, *Stelis*. Centre Suisse de Cartographie de la Faune & Schweizerische Entomologische Gesellschaft, Neuchâtel

Bitsch J, Barbier Y, Gayubo SF, et al (1997) Hyménoptères Sphecidae d'Europe occidentale, Volume 2. Faune de France, 82

Bitsch J, Dollfuss H, Bouček ZRJ, et al (2007) Hyménoptères Sphecidae d'Europe occidentale, Volume 3. Faune de France, 86

Bitsch J, Leclercq J (1993) Hyménoptères Sphecidae d'Europe occidentale, Volume 1. Faune de France, 79

Cariveau DP, Nayak GK, Bartomeus I, et al (2016) The allometry of bee proboscis length and its uses in ecology. PLoS ONE 11:1-13

Ebmer AW (1969) Die bienen des genus *Halictus* Latr. s.l. im Grosraum von Linz (Hymenoptera Apidae). Systematik, biogeographie, ökologie und biologie mit berücksichtigung aller bisher aus Mitteleuropa bekannten Arten. Teile I-III. Naturkundliches Jahrbuch der Stadt Linz 15:133-183

von Hagen E (1994) Hummeln: bestimmen, ansiedeln, vermehren, schützen. Naturbuch Vlg., Augsburg

Scheuchl E, Willner W (2016) Taschenlexicon der Wildbienen Mitteleuropas. Quelle & Meyer
Verlag, Wiebelsheim

Schmid-Egger C, Scheuchl E (1997) Illustrierte bestimmungstabellen der wilbienen Deutschlands un
Österreichs, band 3 – Andrenidae. Norbert Präbst Satz & Druck